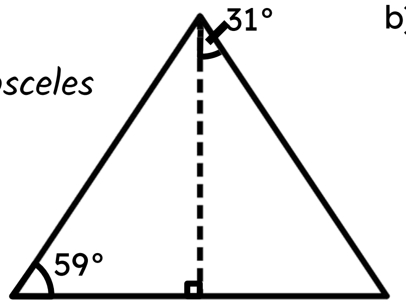
Diagonals of a rhombus

Task A: Isosceles triangles

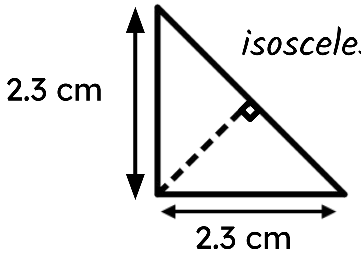
1) Decide which of the following are isosceles triangles. If not, state why not.

a)



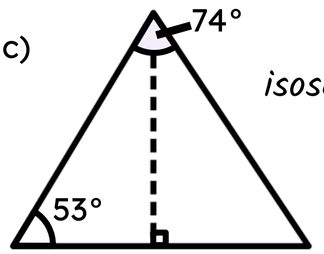
isosceles

b)



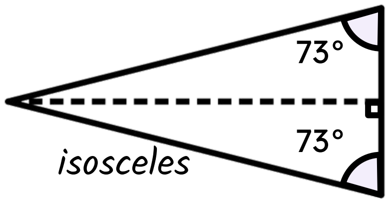
isosceles

c)



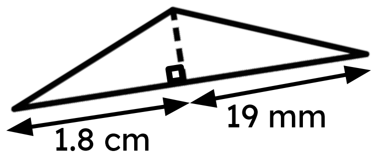
isosceles

d)



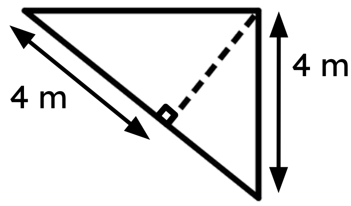
isosceles

e)



Not an isosceles; the altitude does not bisect the base edge.

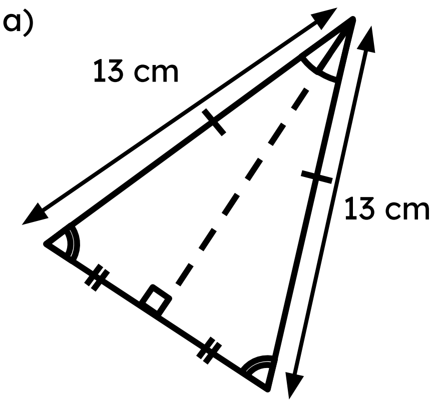
f)



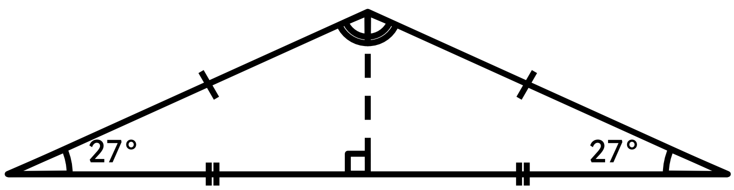
Not an isosceles; there are not two equal edges

2) Add notation marks to the diagrams, where appropriate.

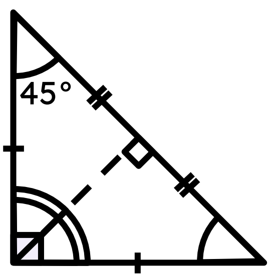
a)



b)



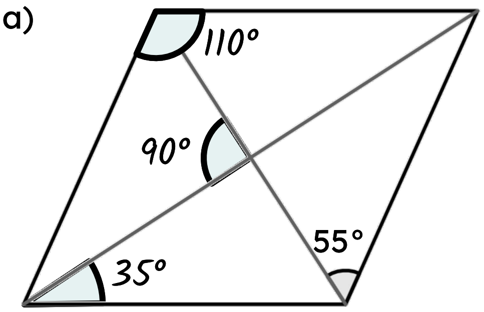
c)



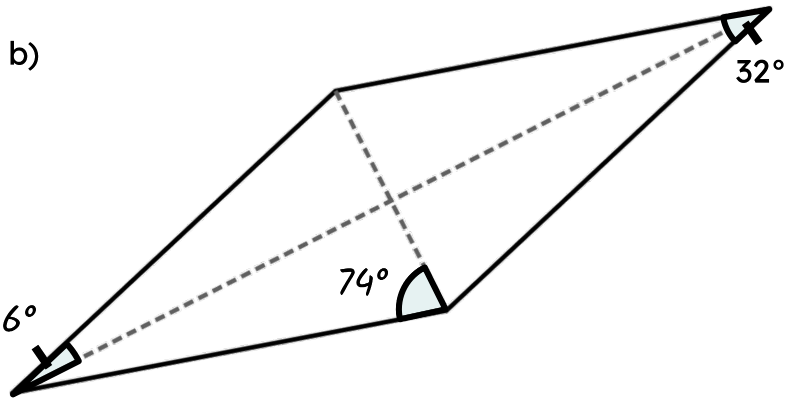
Task B: Properties of a rhombus

1) Work out the missing angles in these rhombi

a)



b)



2) Work out the area of the rhombus in each part:

a) One diagonal is 8 cm and the other is 6 cm

$24\ cm^2$ E.g. $(0.5 \times 4 \times 3) \times 4$

b) One diagonal is 9 cm and the other is 4.5 cm

$20.25\ cm^2$ E.g. $(0.5 \times 4.5 \times 4.5) \times 2$

c) Half of one of the diagonals is 5 cm and the other diagonal is 1 cm

$5\ cm^2$ E.g. $(0.5 \times 5 \times 1) \times 2$

3) Construct a rhombus from this set of perpendicular lines.

An example: